Mana Rasta

మన రాస్తా · *"Our Road"*

AI-powered civic pothole reporting for Indian metros — closing the loop between citizens and municipal engineers.

300	4M+	22K+
GHMC WARDS	CITIZENS SERVED	POTHOLES / YEAR

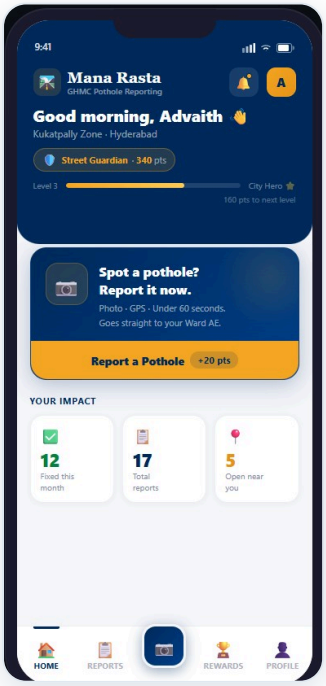
THE PROBLEM

Hyderabad residents have no reliable way to report potholes that actually get fixed. Existing portals get flooded with duplicate, low-quality, or unverifiable reports — and GHMC engineers waste hours triaging noise instead of repairing roads. Citizens never hear back, and repeat reports pile up.

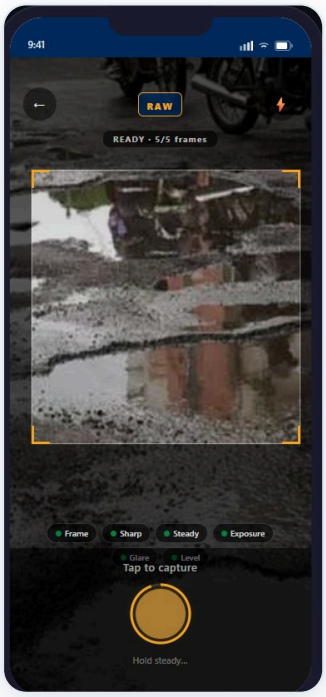
THE SOLUTION

A Full-Stack Civic Infrastructure Platform

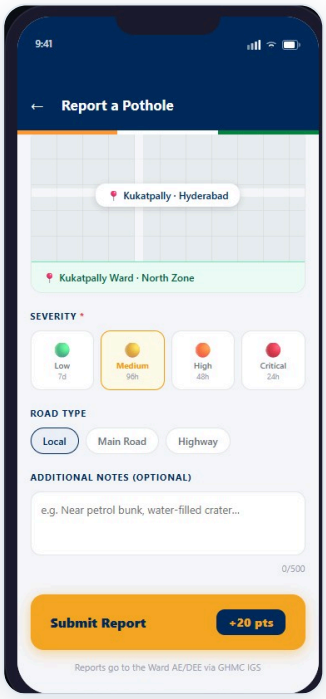
Mana Rasta is an end-to-end system: a citizen mobile app, an AI capture module that validates report quality at the moment of submission, a real-time admin dashboard for GHMC engineers, and a fraud detection layer that keeps the data clean. Every component is production-ready and already built.



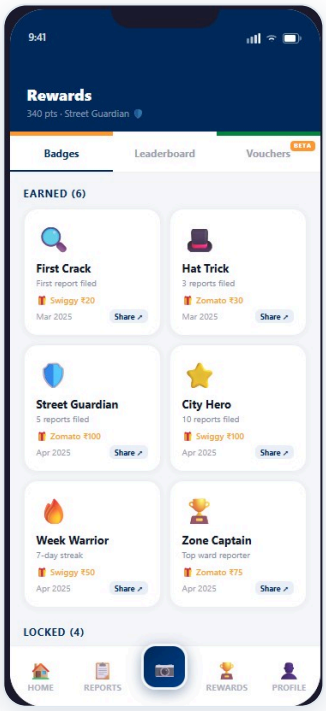
HOME SCREEN



AI CAPTURE



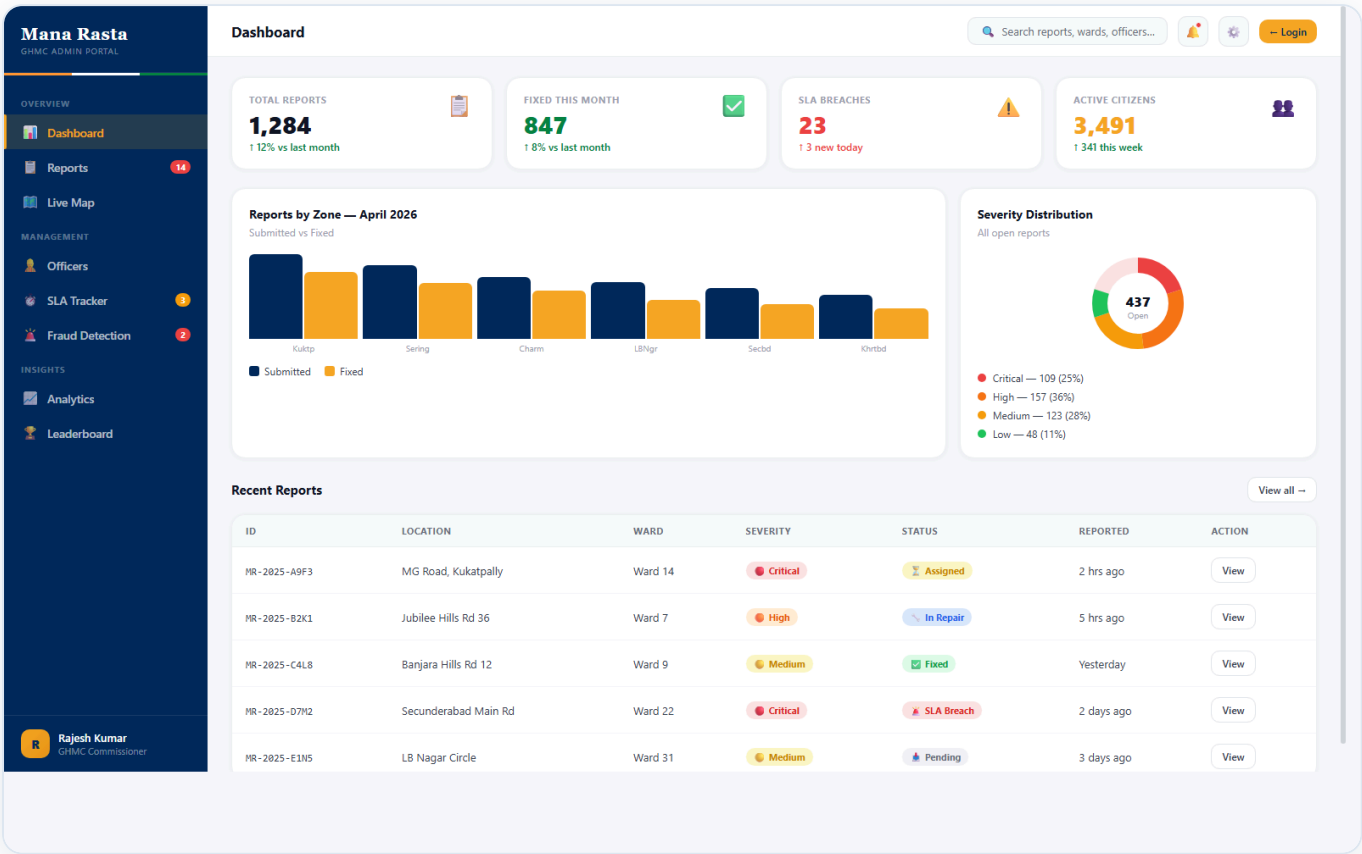
REPORT FLOW



REWARDS

ADMIN DASHBOARD

A four-tier portal — Ward AE — Circle DEE — Zone EE — Commissioner — surfaces exactly what each level needs to act. Ward engineers see a live report queue, priority-sorted by severity. Circle and Zone officers track SLA breaches, repeat complaints, and resolution rates across their jurisdiction. The Commissioner sees the city. Every layer accountable, no report lost.



FEATURES

AI Capture Quality Gate

Real-time Laplacian sharpness, exposure, and pothole candidate scoring — blocks bad images before submission.

10-Layer Fraud Prevention

IP/device rate limits, impossible travel detection, perceptual hash dedup, and auto-ban at risk score ≥ 80.

PostGIS Ward Resolution

Every report is spatially resolved to GHMC ward boundaries for accurate routing to the right engineer.

5-Tier Gamification

Five report-count tiers — from Pothole Spotter to Rasta Legend — with Swiggy, Zomato and Amazon Pay vouchers driving repeat engagement.

BUSINESS MODEL

Hyperlocal Advertising

Tyre brands, auto service, vehicle insurance — ads served at ward level to users with demonstrated road condition engagement.

Government Licensing

GHMC is the pilot. Same infrastructure, swappable ward polygon dataset. Chennai, Pune, Bengaluru — recurring municipal contracts.

Data Partnerships

Verified ML-classified pothole dataset at ward-level granularity — sellable to Google Maps, Waze, and urban planning bodies.

TECH STACK

Mobile	React Native · Expo Router v3
Backend	Node.js · Express · Redis 7
Database	PostgreSQL 16 · PostGIS 3
ML / AI	Python FastAPI · PyTorch · OpenCV
Dashboard	React · Vite · Tailwind CSS
Infra	Docker Compose · S3-compatible

KEY DIFFERENTIATORS

- Quality gate before submission — not after
- Spatial routing to exact GHMC ward engineer
- Fraud detection built into core architecture
- Full citizen feedback loop closes the accountability gap

AI QUALITY PIPELINE

Identity Check
IP throttle · device fingerprint · rate limits

Location Check
Geo boundary · impossible travel · GPS tag

Capture Validation
Blur · brightness · duplicate pHash (OpenCV)

Classification
PyTorch model identifies & scores pothole severity

High-Quality Reports
Only verified potholes reach GHMC engineers

PROJECTED METRICS (YEAR 1 PILOT)

1.35L
MAU — 1% OF 300 WARDS (EST.)

20,000+
REPORTS (YEAR 1)

40%
REPEAT REPORTERS (EST.)

₹500Cr+
ADDRESSABLE MARKET (INDIA)

₹1.6Cr
TOTAL REVENUE (LICENSING + ADS) (EST.)